So if we plug our result in (5) into eq-n (1):

now by (3)

$$u_{xy} = u_{\xi\xi}\,\xi_x\,\xi_y + u_{\xi\eta}\left(\xi_x\eta_y + \xi_y\eta_x\right) + u_{\eta\eta}\,\eta_x\,\eta_y = 0$$

$$\Rightarrow \quad u_{\xi\xi}\cdot\left(\frac{\sqrt{2}}{2}\right)\cdot\left(\frac{\sqrt{2}}{2}\right) + u_{\xi\eta}\underbrace{\left\{\left(\frac{\sqrt{2}}{2}\right)\cdot\left(\frac{\sqrt{2}}{2}\right) + \left(\frac{\sqrt{2}}{2}\right)\cdot\left(-\frac{\sqrt{2}}{2}\right)\right\}}_{=0} + u_{\eta\eta}\cdot\left(-\frac{\sqrt{2}}{2}\right)\cdot\left(\frac{\sqrt{2}}{2}\right) = 0$$

$$\Rightarrow \quad \frac{1}{2}u_{\xi\xi} - \frac{1}{2}\cdot u_{\eta\eta} = 0 \quad \Big\}\cdot 2 \text{ Both sides (BS):}$$

$$\boxed{u_{\xi\xi} - u_{\eta\eta} = 0}$$ establishing the desired result. ∎

Prob 4. (ZT, Ch. 5, 7.7)

Describe the regions in the (x,y) plane where the ↓ PDE's are hyperbolic, parabolic, or elliptic:

(a) $2u_{xx} + 4u_{xy} + 3u_{yy} - u = 0$

(b) $y\,u_{xx} - 2u_{xy} + e^{x}u_{yy} + x^2 u_x - x = 0$

(c) $y\,u_{xx} + 2x\,u_{xy} + x\,u_{yy} + 3u_x - 4y\,u_y + e^{x}u = 0.$

Sol-n:

(a) given the PDE

(1) $$\overset{a}{2}\,u_{xx} + \overset{b}{4}\,u_{xy} + \overset{c}{3}u_{yy} - u = 0$$

By classification of 2nd order linear PDE's:

(2) $$\begin{cases} \Delta = b^2 - 4ac > 0 & \text{the eq-n is hyperbolic} \\ \Delta = b^2 - 4ac = 0 & \text{the eq-n is parabolic} \\ \Delta = b^2 - 4ac < 0 & \text{... elliptic.} \end{cases}$$

So for (1)

$\Delta = 4^2 - 4\cdot(3)\cdot(2) = 16 - 24 = -8 < 0 \Rightarrow$ hence,

the eq-n in (1) is <u>elliptic</u>. ∎

(b) $\overset{a}{y}\, u_{xx} \overset{b}{-2} u_{xy} + \overset{c}{e^x}\, u_{yy} + x^2 u_x - u = 0$

• $\Delta = b^2 - 4ac = (-2)^2 - 4(y)\cdot(e^x) = 4 - 4ye^x = 0$ (For parab-c)

(3) $4ye^x = 4 \overset{e^x \neq 0}{\underset{\text{so}}{\Rightarrow}} \boxed{y = \frac{1}{e^x} = \left(\frac{1}{e}\right)^x}$ ← Parabolic condition

• For being hyperbolic $4 - 4ye^x > 0 \Rightarrow 4ye^x < 4 \Rightarrow$

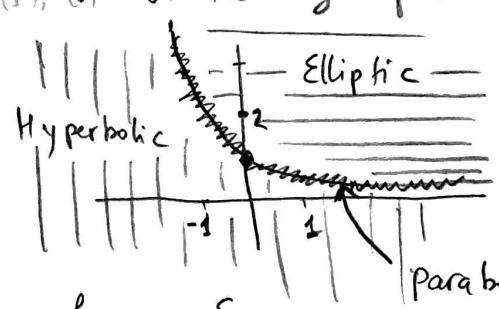
(4) $\Rightarrow \boxed{y < \left(\frac{1}{e}\right)^x}$ ← Hyperbolic condition

• For being elliptic

$\Delta = b^2 - 4ac = 4 - 4ye^x < 0 \Rightarrow 4ye^x > 4 \Rightarrow$

(5) $\Rightarrow \boxed{y > \left(\frac{1}{e}\right)^x}$ ← Elliptic condition.

Putting (4), (5), (6) on the xy plane;



Parabolic on $y = \left(\frac{1}{e}\right)^x$.

(6) (c) $\overset{a}{y}\, u_{xx} + \overset{b}{2x}\, u_{xy} + \overset{c}{x}\, u_{yy} + 3u_x + \ldots = 0$

(7) $\Delta = b^2 - 4ac = (2x)^2 - 4(y)\cdot(x) = 4x^2 - 4yx$

• For Parabolic: $\Delta = 0$

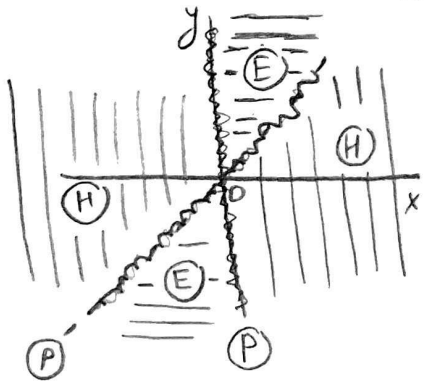
(8) $\Rightarrow 4x^2 - 4yx = 0 \Rightarrow 4x(x-y) = 0 \Rightarrow \boxed{x = 0}$ (P) or $\boxed{y = x}$ (P)

• For Hyperbolic: $\Delta > 0$

$\Rightarrow 4x^2 - 4yx > 0 \Rightarrow 4x(x-y) > 0$ $\boxed{x > 0 \ \& \ y < x}$ (H) or $\boxed{x < 0 \ \& \ y > x}$ (H)

• For Elliptic, $\Delta < 0$: $\Rightarrow$

$\Rightarrow \Delta = 4x^2 - 4yx < 0 \Rightarrow 4x(x-y) < 0$ $\quad$ (E) $\boxed{x < 0 \ \& \ y < x}$ or (E) $\boxed{x > 0 \ \& \ y > x}$



Prob 5. (2 & T, Ch. 5, 5.1)

Let P be the nonlinear operator in $\mathbb{R}^2$ defined by

(1) $$P(u) = \left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}\right)^2 - u^2.$$

Show that each of the ↓ f-ns

(2) $$u_1(x,y) = e^x, \quad u_2(x,y) = e^{-y}$$

are sol-ns of the homogenous eq-n

(3) $$P(u) = 0$$

while their sum

(4) $$u(x,y) = e^x + e^{-y}$$ is not a sol-n.

Does this violate the principle of superpos-n?

Sol-n: [Preamble].

$$P(u_1) = \left(\frac{\partial}{\partial x}[e^x] + \frac{\partial}{\partial y}[e^x]\right)^2 - [e^x]^2 = 0 \quad \text{by (1) \& (3).}$$

$$\Rightarrow (e^x + 0)^2 - e^{2x} = e^{2x} - e^{2x} \equiv 0 \checkmark, \text{ so } u_1 = e^x$$

is a sol-n for (2).

Similarly,